E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 13-A

Linear spatial transformations

Berkeley ENGINEERING

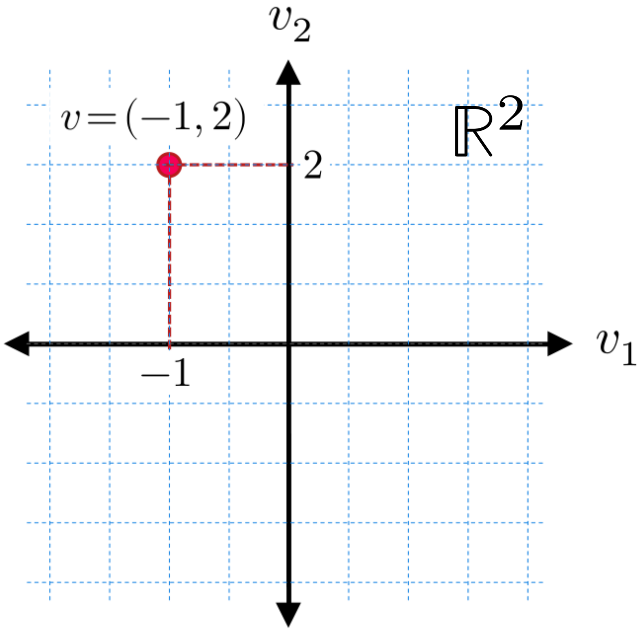
Matrix multiplication in NumPy

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} =$$

see lecture notebook for code...

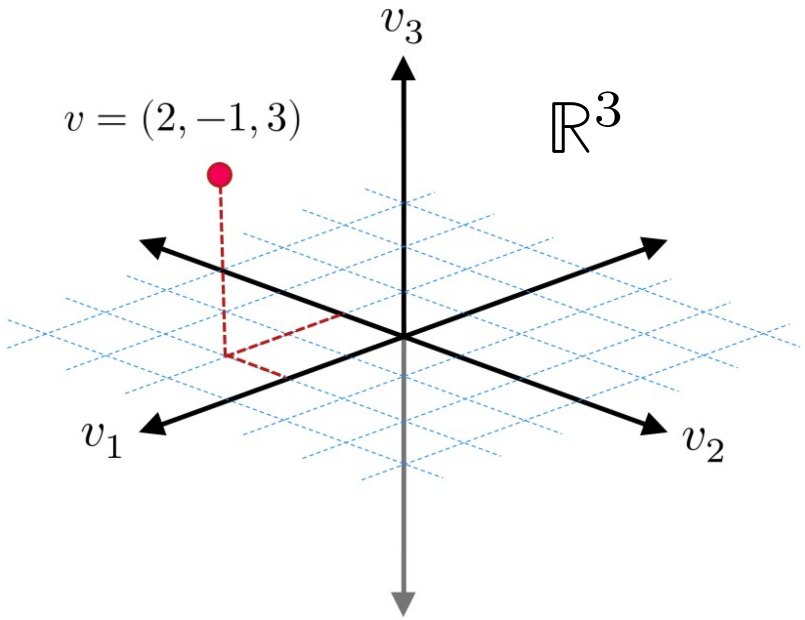
2D vector space

$$v = (v_1, v_2) \in \mathbb{R}^2$$



3D vector space

$$v = (v_1, v_2, v_3) \in \mathbb{R}^3$$



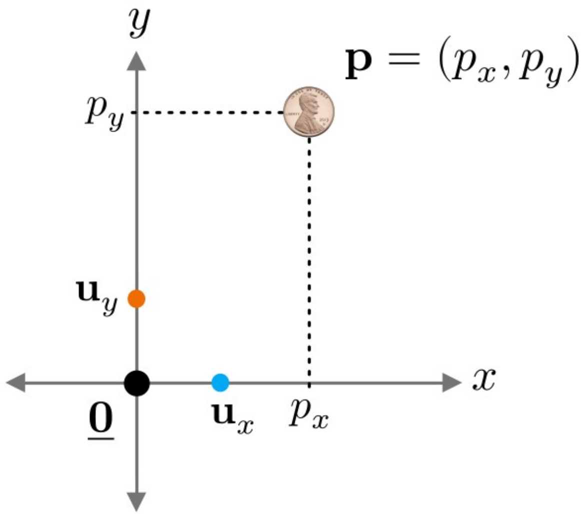
Cartesian coordinate system

- A Cartesian coordinate system is a tool for describing the positions of objects in space.
- It consists of several landmarks: the ***origin*** and several ***unit vectors***.
- The line segments between the origin and each of the unit vectors must form **right angles**.

● $\underline{\mathbf{0}} = (0, 0)$ … origin

● $\mathbf{u}_x = (1, 0)$ … unit vector

● $\mathbf{u}_y = (0, 1)$ … unit vector



Coordinate transformation

A ***coordinate transformation*** is a function that maps the representation of an object in one coordinate system to its representation in another.

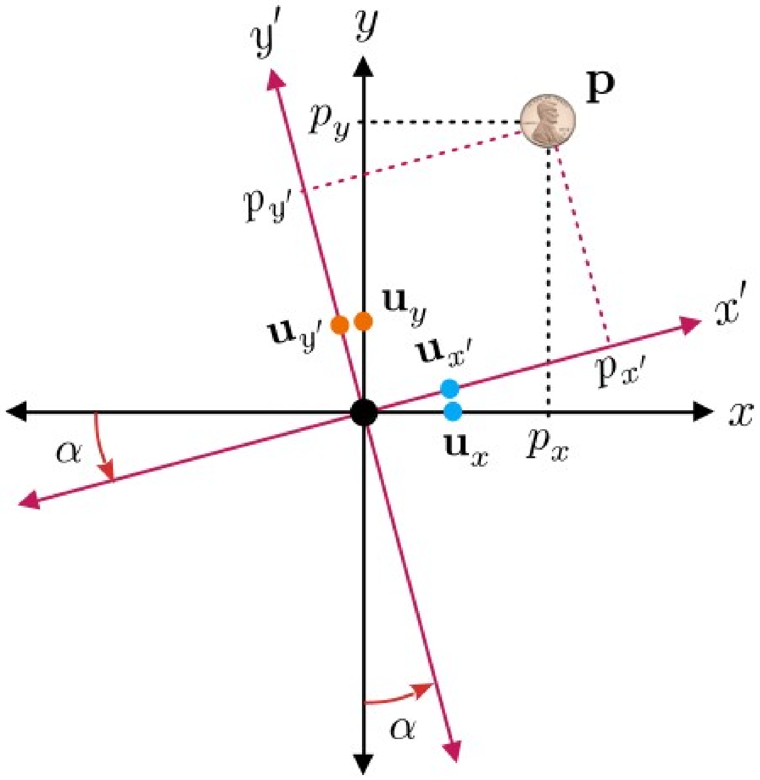
Example: 2D rotation

- Two coordinate systems with the same origin.

$$C_1 = (\underline{\mathbf{0}}, \mathbf{u}_x, \mathbf{u}_y)$$
$$C_2 = (\underline{\mathbf{0}}, \mathbf{u}_{x'}, \mathbf{u}_{y'})$$

- The same point has different representations in C1 and C2.

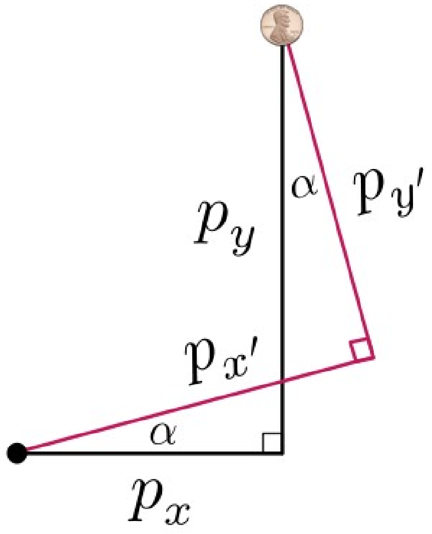
$$\mathbf{p} = (p_x, p_y) \quad \ldots C_1$$
$$\mathbf{p} = (p_{x'}, p_{y'}) \quad \ldots C_2$$



- Coordinate transformation: $$T : \mathbb{R}^2 \to \mathbb{R}^2$$ $$(p_x, p_y) \mapsto (p_{x'}, p_{y'})$$

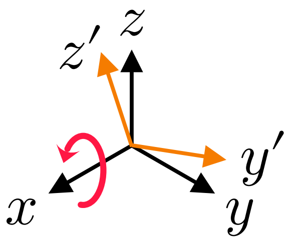
- T is a *linear transformation*, i.e. a *matrix multiplication*.

$$\begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$



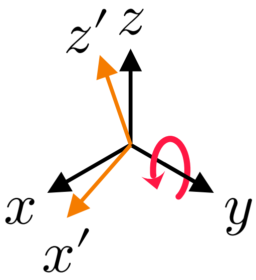
3D axis-aligned rotations

$\underline{x \text{ axis}}$:



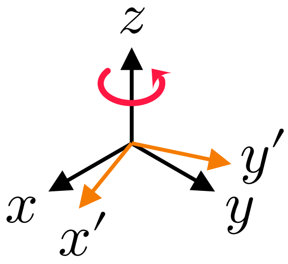
$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) \\ 0 & -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

$\underline{y \text{ axis}}$:



$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & 0 & -\sin(\alpha) \\ 0 & 1 & 0 \\ \sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

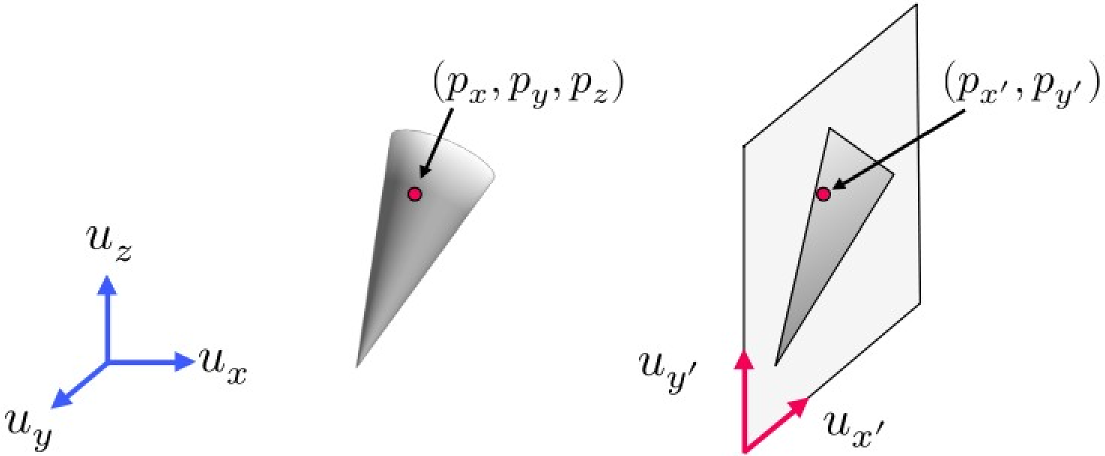
$\underline{z \text{ axis}}$:



$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

3D→2D Projections

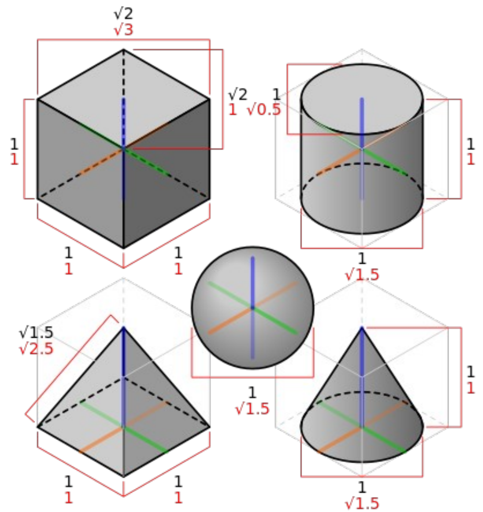
- A **projection** is a coordinate transformation from 3D to 2D.
- Assume that the origin of the two coordinate systems are the same.

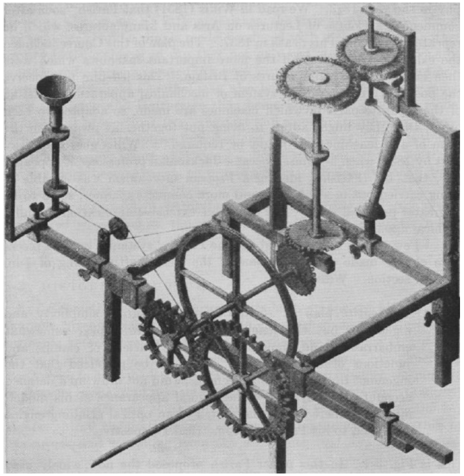


$$\begin{bmatrix} p_{x'} \\ p_{y'} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

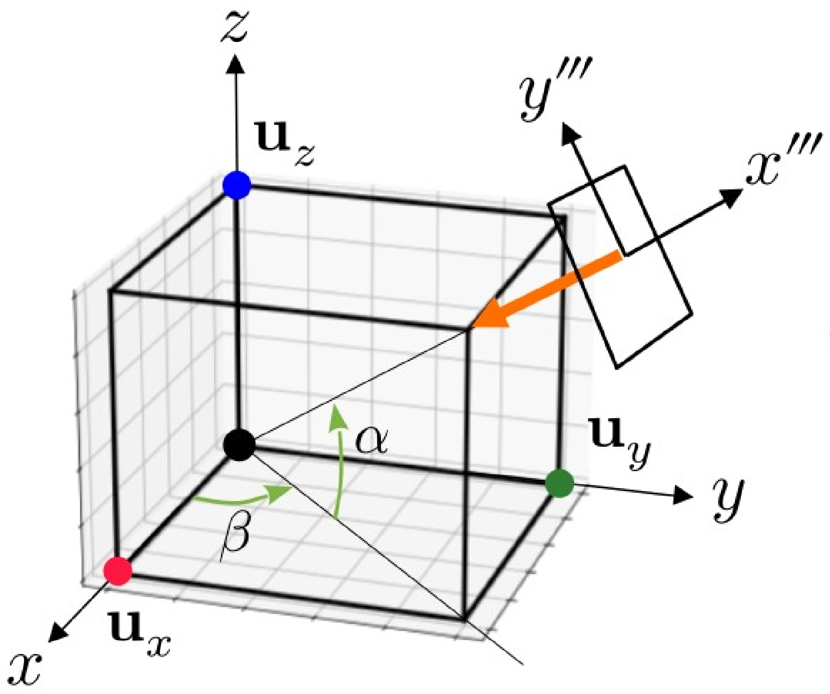
Isometric projections

An **isometric projection** is a type of 3D→ 2D projection that preserve relative lengths.

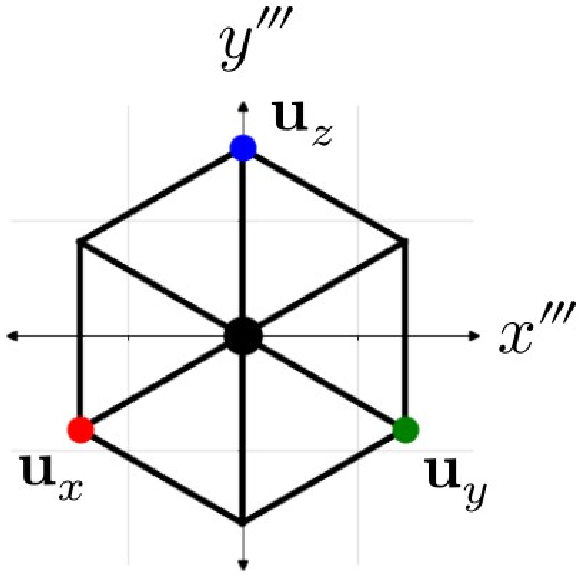




World coordinates



Isometric projection



$$\beta = 45°$$

$$\alpha = \text{atan}\left(1/\sqrt{2}\right) \approx 35.264°$$

Isometric projection as a linear transformation

1. Rotation of β about z axis. $(x, y, z) \rightarrow (x', y', z')$

$$
\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \cos(\beta) & \sin(\beta) & 0 \\ -\sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}
$$

$$
= \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_1} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}
$$

2. Rotation of $-\alpha$ about y' axis. $(x', y', z') \rightarrow (x'', y'', z'')$

$$\begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & 0 & \sin(\alpha) \\ 0 & 1 & 0 \\ -\sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} \\ 0 & 1 & 0 \\ -1/\sqrt{3} & 0 & \sqrt{2}/\sqrt{3} \end{bmatrix}}_{R_2} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix}$$

3. Projection: $y'' \rightarrow x''' \; , \; z'' \rightarrow y'''$

$$\begin{bmatrix} p_{x'''} \\ p_{y'''} \end{bmatrix} = \underbrace{\begin{bmatrix} & & \\ & & \end{bmatrix}}_{R_3} \begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix}$$

Put them together:

$$\begin{bmatrix} p_{x'''} \\ p_{y'''} \end{bmatrix} = R_3 \begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix} = R_3 R_2 \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \underbrace{R_3 R_2 R_1}_{M} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$